

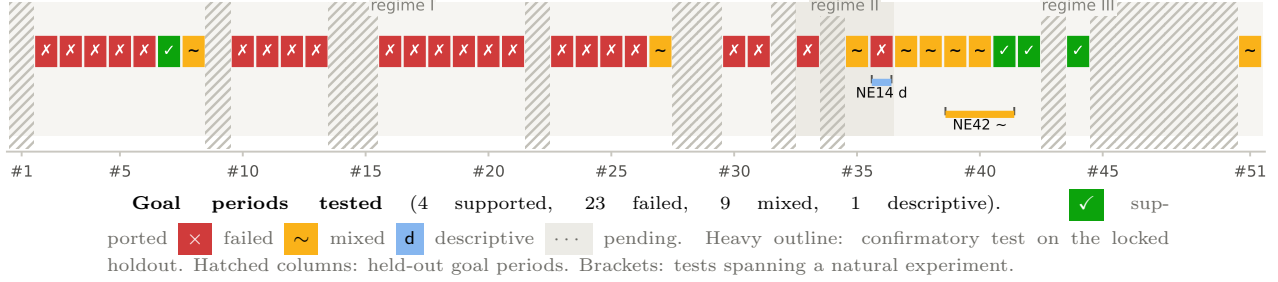
H67: A lagged criticality dial

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 3/5 Usefulness 2.5/5

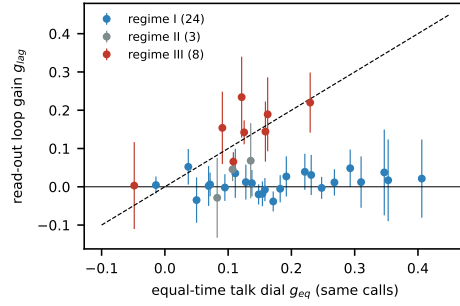
Bottom line. Read-out talk coupling is small and subcritical ($g_{lag} \approx 0.13$ in regime III, matching the Fano-implied 0.149 [0.123, 0.175]; ≈ 0 in regime I on either clock). H50's twice-larger value is an artifact of its estimator design.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H25's equal-time talk dial measures feedback within one minute. Coupling acts at the recipient's next model call, so HH262 asked whether a dial built on that read-out call shows larger gains that are still subcritical. Round 1 covered 66 non-reserved units in 35 periods, with 1.47M receiving calls. The read-out loop gain g_{lag} counts the extra talk messages that one message causes through reading. Every unit is subcritical (all upper bounds < 0.40). The gain is 0.13 in regime III and ≈ 0 elsewhere, and the **HH claim (lagged > equal-time) fails** (7/35 periods). Round 2 located H50's factor 2. In regime III it comes from H50's pair-RD design; in regime I it is call-class composition. H111's fluctuations give an independent total gain of 0.149 [0.123, 0.175], which matches g_{lag} . Regime I shows no gain on its chat clock either.



Observables. Round 1. Each point is one goal period (random-effects pool of its units, 95% CI). The x-axis is H25's equal-time talk dial on the same calls, the y-axis the read-out loop gain. Regime-I periods sit on $g_{lag} \approx 0$ while g_{eq} reads 0.05–0.40: a fast shared field, not read-out coupling. Regime-III periods lie near the diagonal at 0.07–0.23.

PHYSICAL MODEL

Linear response on the call clock (models 01, 09). Agent i 's call c is a talk call ($Y_{ic} = 1$) or not. R_{ic}^m counts peer messages read at c that were posted within the call's latency d_c before its start. P_{ic} counts messages posted within d_c after the start (in flight, unreadable). $a_{i,d,\kappa}$ is an agent $\times$ day $\times$ call-class field; $\bar{r}$ is the readers per message and $\bar{m}$ the messages per talk call. Round 2 adds the Fano-implied gain: H111's collective Fano ratio Φ inverted through its sum rule.

$$P(Y_{ic}=1) = a_{i,d,\kappa} + \beta_m R_{ic}^m + \beta_o R_{ic}^o + \beta_P P_{ic} + \gamma \cdot E_{ic} + \rho Y_{i,c-1}$$

$$J_1^* = \beta_m - \beta_P, \quad g_{lag} = \bar{m} \bar{r} J_1^*, \quad T/T_c = 1/g_{lag}$$

$$G_H = \bar{m} \bar{r} \sum_{h \leq H} \delta(h), \quad \delta(h) = \sum_{k < h} [\beta(R_{c-k}^m) - \beta(P_{c-k})]$$

$$\Phi_{obs}(10 \text{ min}) = \Phi_{pred}(g_{Fano}) \approx (1 - g_{Fano})^{-2}$$

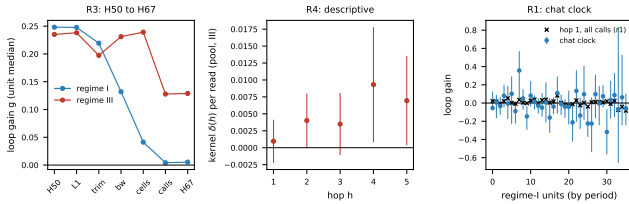
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: report g_{lag} , not the equal-time dial or a pair-RD jump, as distance to a talk cascade. One regime-III talk message causes about 0.13 further messages through reading (amplification $\times 1.15$, $T/T_c \approx 8$). The fluctuation bound caps the full gain, all hops included, at 0.175. Naming carries the coupling: a named message raises the next-call talk probability by 0.08, an unnamed one by 0.004. In regime I, talk is set by the chat-turn schedule, not by reading. No village configuration came near runaway, so this is a gauge, not a lever.

H67-lagged-criticality-dial · generated 2026-10-04 20:20 by infra/summaries/build_summaries.py · card hypotheses/H67-lagged-criticality-dial/README.md · ratings: H67 agent (suggested, round 2)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 subcritical every-where	max upper bound 0.39; all units < 0.5	supported
P2 $g_{lag} > g_{eq}$ in $\geq 2/3$	7/35 periods; median ratio 0.09	failed
P3 rank-consistent with H42	$\rho = 0.55$ ($p = 0.0007$)	supported
P5 named $\geq 5 \times$ named (III)	un- 0.079 vs 0.004 ($\times 19$)	supported
P7 J_1^* falls with N	$\rho = -0.59$; g_{lag} flat	supported
<i>Round 2 (2026-10-05)</i>		
R3-P1 reproduce H50	L1/L0 0.94, ρ 0.87	supported
R3-P2 gap in cells + placebo (III)	gap in pair RD $\rightarrow$ call counts (share 1.0); placebo 0.016	failed
R3-P3 first stage $\Delta R \approx 1$	0.04	supported
R3-P4 cells remove $\geq 70\%$ (I)	69% (85–95% other readouts)	failed narrowly
R4 multi-hop G_5	synthetic failed; pool 0.24 $[-0.03, 0.51]$	descriptive
R4-P4 Fano check	g_{Fano} 0.149 $[0.123, 0.175]$ vs g_{lag} 0.127	hop closes 1
R1-P1 chat-clock (I) ≥ 0.05	gain -0.022 $[-0.052, 0.009]$	failed
R1-P2 explains excess	H111 r_F 1.35 unchanged	failed



Second figure. Round 2. Left: unit-median gain along the ladder from H50's estimator to H67's. Regime I collapses at the call-class cells; regime III drops only at the switch from pair RD to per-call counts. Middle: multi-hop kernel $\delta(h)$, regime-III pool (descriptive; the design failed its synthetic test). Right: regime-I units on the chat clock (points, 95% CI) and round 1's hop-1 gain (crosses). Both sit at 0.

GOAL PERIODS IN DETAIL

G02 $\times$ failed (round 2, chat clock; round 1: failed)
G03 $\times$ failed (round 2, chat clock; round 1: failed)
G04 $\times$ failed (round 2, chat clock; round 1: failed)
G05 $\times$ failed (round 2, chat clock; round 1: supported)
G06 $\times$ failed (round 2, chat clock; round 1: failed)
G07 $\checkmark$ supported (round 2, chat clock; round 1: failed)
G08 $\sim$ mixed (round 2, chat clock; round 1: mixed)
G10 $\times$ failed (round 2, chat clock; round 1: failed)
G11 $\times$ failed (round 2, chat clock; round 1: failed)
G12 $\times$ failed (round 2, chat clock; round 1: failed)
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	2	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The multi-hop design is too noisy per unit (SD 0.4–1.5 in synthetic worlds) and cannot resolve hop 1 in real data. The full-gain bound rests on H111's linear sum rule. Start-time error attenuates the chat-clock gain to about 0.23 of the truth, so regime I is bounded at < 0.04 , not zero. Post hoc: without conditioning on call mode, regime I shows 0.035 $[0.023, 0.046]$ above a call-skeleton null; 3/4 of the "read $\rightarrow$ chat-mode" effect is the chat-turn schedule. The unnamed part sits at the placebo's resolution. Predictions were written knowing H25, H42, H50 and H111.

ROUND 2

Confirm $g_{lag} \approx g_{Fano}$ on the reserved regime-III units. Identify the pair-sampling term (0.18 vs 0.13). Test the chat-turn schedule as the regime-I fast field behind H111's excess.

- **Essence.** a lag-aware, field-free loop gain that an operator can trust as "distance to runaway talk", on the clock the scaffold imposes.
- **H67-R1** Regime-I clock.
- **H67-R2** Named vs unnamed on the holdout.
- **H67-R3** Reconcile with H50.
- **H67-R4** Multi-generation gain.